Vision-Language Model Purified Semisupervised Semantic Segmentation for Remote Sensing Images

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***Abstract*—The semisupervised semantic segmentation (S4) can learn rich visual knowledge from low-cost, unlabeled images. However, traditional S4 architectures all face the challenge of low-quality pseudolabels, especially in the teacher–student framework. We propose a novel SemiEarth model that introduces vision-language models (VLMs) to address the S4 issues in the remote sensing (RS) domain. In particular, we invent a VLM pseudolabel purifying (VLM-PP) structure to purify the teacher network's pseudolabels, achieving substantial improvements. Especially in multiclass boundary regions of RS images, the VLM-PP module can significantly improve the quality of pseudolabels generated by the teacher, thereby correctly guiding the student model's learning. Moreover, since VLM-PP equips VLMs with open-world capabilities and is independent of the S4 architecture, it can correct mispredicted categories in low-confidence pseudolabels whenever a discrepancy arises between its prediction and the pseudolabel. We conducted extensive experiments on multiple RS datasets, which demonstrate that our semiEarth achieves SOTA performance. More importantly, unlike previous SOTA RS S4 methods, our model not only achieves excellent performance but also offers good interpretability. The code is released at https://github.com/wangshanwen001/SemiEarth**

***Index Terms*—Remote sensing (RS), semisupervised semantic segmentation (S4), vision-language model (VLM).**

I. Introduction

REMOTE sensing (RS) image semantic segmentation is a core technology for understanding surface cover and monitoring environmental changes [1], [2], [3], [4], but it heavily relies on large-scale annotated data [5], [6]. At the same time, enormous unannotated RS images are acquired from satellites and drones, which are an unexplored treasure for understanding the Earth. Semisupervised semantic segmentation (S4) can significantly reduce annotation requirements, which is an effective technique for such data. In RS scenarios, S4 specifically enhances the generalization ability in practical applications such as farmland monitoring and disaster assessment with limited labeled data [7], [8], [9]. So far, S4 research primarily focuses on efficient pseudolabel generation [10], consistency regularization [11], and self-training strategies [12] to mine the potential information of unlabeled data. However, traditional S4 architecture cannot address the challenges posed by low-quality pseudolabels. In particular, teacher–student architectures suffer from low-quality pseudolabels generated by the teacher network, which can mislead the student model [11]. Self-training strategies [13] and consistency regularization methods like FixMatch [14], which rely on a single network, also produce unreliable pseudolabels. These pseudolabels, generated by either a teacher or the model itself, are the key to model training. However, these incorrect pseudo-labels often lead the model to unstable, erroneous results [15], [16]. Even though researchers have recently focused on solving this problem, they still struggle with the drawbacks of common S4 architectures. For example, several works attempt to filter pseudolabels based on confidence and uncertainty [11], [17], [18], but the selected samples still contain a large amount of noise. It is noteworthy that the low-confidence regions filtered out by confidence and uncertainty are located at ambiguous multiclass boundaries. It will make the model ineffective if we exclude these pixels from training. Moreover, incorrect pseudolabels and training errors will be amplified during training, especially in multiclass boundary areas that require fine segmentation.

To address these issues, we innovatively introduce visual language models (VLMs) into the RS S4 domain, proposing a novel semisupervised RS model called SemiEarth. SemiEarth abandons various complex mechanisms, such as multiple perturbations and various weighting schemes [10]. Instead, it introduces VLMs to purify low-quality pseudolabels. The schematic structures of SemiEarth and mainstream S4 methods are shown in Fig. 1. Unlike traditional S4 architectures that focus on perturbation and consistency, SemiEarth establishes a novel scheme that integrates visual-linguistic features. We introduce the VLM pseudolabel purifying (VLM-PP) structure for SemiEarth to purify low-quality pseudolabels generated by the teacher model. Rather than simply discarding low-confidence pseudolabels, VLM-PP seeks to refine and enhance the quality of those with extremely low confidence. More importantly, the VLM-PP can rectify errors when the teacher model generates low-quality pseudolabels with mis-

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classifications. The main contributions of this article are as follows.

1) We propose a novel semisupervised model, SemiEarth, which for the first time introduces VLMs into the RS S4 domain to address the challenge of low-quality pseudolabels generated by teacher models.
2) We propose a VLM-PP module to purify low-quality pseudo-labels from the teacher, significantly enhancing pseudolabel quality and effectively preventing the student from being misled.
3) We propose a rectification mechanism for false pseudolabels in the RS S4 domain, leveraging the VLM's independent judgment of low-quality pseudolabels outside the common RS S4 framework.
4) Extensive experiments on RS datasets demonstrate that our model significantly outperforms traditional SOTA methods and offers very good interpretability.

The rest of this article is organized as follows. Section II briefly introduces the related work. In Section III, our model, SemiEarth, is proposed and discussed. Section IV shows the experimental results and compares them with SOTA methods. Section V concludes this article.

II. Related Work

In recent years, with the maturity of deep learning, an increasing number of S4 models and VLMs have been applied in the field of RS. This section systematically reviews recent advances in S4 models and VLMs for RS, providing the necessary background for our proposed model.

A. Semisupervised Semantic Segmentation

The advent of deep learning has revolutionized the RS domain, with fully supervised networks like the FCN [19], SegNet [20], and U-Net [21] achieving remarkable progress in semantic segmentation. These models have promising segmentation capabilities and stability, but their success hinges on large-scale, high-quality labeled datasets, which are costly and labor-intensive to acquire. On the contrary, RS S4 models aim to leverage abundant unlabeled data instead of well-labeled data, thereby reducing dependence on expensive pixel-level annotations [22], [23], [24]. The recent RS S4 models primarily focus on two aspects of innovation. One is to address the inherent limitations of the S4 algorithms themselves [10], [25]. For example, previous studies by Lu et al. [18], Huang et al. [26], and Chen and Zhu [27] have tried to solve the problems of low quality of pseudolabels and the inherent distribution mismatch between data. However, they targeted S4 structure improvement and mismatched the inherent characteristics of RS images, such as multiscale information and the complex features of multiclass boundary regions. The other is to target the unique characteristics of RS images. For instance, Wang et al. [11], Ni et al. [28], and Xin et al. [29] focused on addressing multiscale features and high interclass similarity specific to RS images. In particular, Wang et al. [11] and Ni et al. [28] addressed the multiscale challenges by leveraging label space for contextual label readjustment and employing multiscale uncertainty consistency, respectively. Facing the high interclass similarity in RS images, Wang et al. [11] and Xin et al. [29] adopted contrastive learning and cross-teacher–student attention networks. However, they designed quite complex structures and made limited performance improvements. For example, the method by Wang et al. [11] only achieved a marginal improvement of about 1%–2% in mean intersection-over-union (mIoU) compared to the previous SOTAs. In addition, some RS S4 methods, including MCMCNet [30] and SemiRoadExNet [31], pay attention to specific categories of RS without generalizability.

Overall, these RS S4 methods in the RS field are constrained by traditional architectures, failing to accurately remove errors in pseudolabels, especially at multiclass boundary areas requiring fine segmentation of RS images. Particularly, pseudolabels play a significant role in the training procedure. This study introduces a pseudolabel purification module, VLM-PP, which is independent of the S4 architecture. It can effectively purify out low-quality pseudolabels and rectify the teacher's errors, significantly improving the performance of the student.

B. VLMs for RS

The existing deep learning methods in the RS field have primarily focused on visual processing while neglecting semantic understanding [32], [33], [34]. For example, visual models may misclassify building roofs as highways when the visual features of the pixels are similar. The main reason is that these models lack the common-sense knowledge that a highway cannot be on a building's roof. VLMs can jointly reason about images and their textual descriptions, thereby gaining a deep understanding of semantic relationships [35], [36], [37].

In particular, VLMs provide an opportunity to explore the integration of general and expert knowledge into visual analysis tasks for RS data [38], [39]. For instance, VLMs are aware that ships are likely to be located on water rather than on land [40]. Therefore, VLM-based segmentation models often avoid misidentifying categories on the ground as ships, demonstrating a potential improvement for RS analysis. Currently, some studies have explored the application of VLMs in various RS analysis tasks, including RS image captioning (RSIC) [41], [42], [43], text-based RS image retrieval (RSTBIR) [44], [45], RS visual question answering (RS-VQA) [46], [47], [48], referring RS image segmentation (RRSIS) [40], [49], and open-vocabulary RS segmentation (OVRS) [50], [51], [52].

However, the aforementioned applications of VLMs in the RS field are based on large-scale labeled dataset training. Meanwhile, some pioneer works, such as the SegEarth-OV [53] and RSCLIP [54] models, have explored the zero-shot tasks with VLMs in RS. While they have made important contributions, the improvement of performance is limited due to the inherent constraints of the CLIP model, which stems from its pretraining on image–text pairs. The purpose of this study is to integrate the VLMs with the RS S4 framework to fully utilize the unlabeled RS images. We aim to achieve excellent performance using only a small amount of labeled and a large amount of unlabeled RS image data for training. In addition, the proposed SemiEarth does not train the VLM

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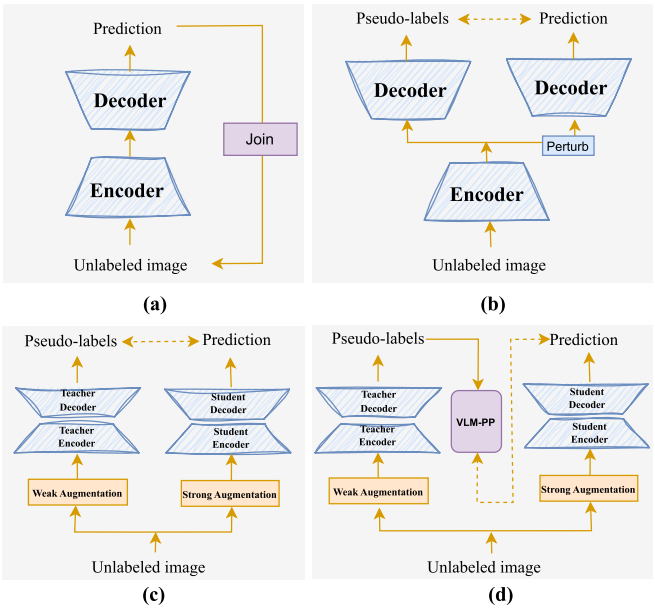


Fig. 1. Structure comparison of S4 frameworks. (a) Self-training semisupervised models, (b) and (c) Consistency regularization models. (d) Our SemiEarth.

model itself but only uses it for the purification of pseudolabels during inference.

III. Method

This section is organized as follows. Section III-A describes our main framework, Section III-B introduces the motivation of the VLM-PP, and Section III-C describes the core principles of the VLM-PP.

A. Preliminaries and Main Framework

Semisupervised learning can train a model with a small amount of labeled data for supervised learning and a large amount of unlabeled data for unsupervised learning. The proposed SemiEarth is a specific semisupervised learning model, as shown in Fig. 2. In particular, we use light grey, light blue, and light beige regions to represent unsupervised learning, supervised learning, and the VLM-PP structure. SemiEarth takes both labeled and unlabeled images as training data for supervised and unsupervised learning, respectively. The VLM-PP is the method to combine and improve these two learning procedures. We first define $D^L = \{(x_i^l, y_i)\}_{i=1}^{N_L}$ as labeled data and $D^U = \{(x_i^u)\}_{i=1}^{N_U}$ as unlabeled. Here $x_i^l \in \mathbb{R}^{H\times W\times 3}$ denotes labeled image, $y_i \in \mathbb{R}^{H\times W\times K}$ is the ground truth, while $x_i^u \in \mathbb{R}^{H\times W\times 3}$ denotes unlabeled image. N_L and N_U are the amount of labeled and unlabeled images, where $N_L << N_U$. H and W specify the height and width of the image. As shown in Fig. 2, the supervised loss $\mathcal{L}_S$ constrains the supervised predictions of the student on labeled data and ground truth. Unsupervised loss $\mathcal{L}_U$ constrains the consistency between students' predictions on unlabeled data and pseudo-labels. The overall loss function $\mathcal{L}$ summarizes supervised $\mathcal{L}_S$ and unsupervised $\mathcal{L}_U$ losses as follows:

$$\mathcal{L} = \mathcal{L}_S + \mathcal{L}_U = \frac{1}{N_L}\sum_{i=1}^{N_L}\mathcal{L}_{CE}\left(p_i^l, y_i\right) + \frac{1}{N_U}\sum_{i=1}^{N_U}\mathcal{L}_{CE}\left(p_i^{u,s}, y_i^{u,t}\right) \quad (1)$$

where p_i^l is the prediction of the labeled image x_i^l, $\mathcal{L}_{CE}$ is cross-entropy loss, $p_i^{u,s}$ is student predictions for unlabeled data x_i^u, and $y_i^{u,t}$ is the pseudolabel.

Similar to most RS S4 architectures, SemiEarth emphasizes unsupervised learning. As shown in the light gray region of Fig. 2, SemiEarth adopts basic teacher–student architecture for unsupervised learning. In particular, the unlabeled data is first applied with weak and strong augmentations, respectively. Weak augmentation provides the teacher with stable features to generate reliable pseudolabels, while strong augmentation enables the student to learn from a broader range of variations. The parameters of the student are updated under the guidance of the pseudolabels generated by the teacher, and the weights θ^t of the teacher model are updated via an exponential moving average (EMA) of the student's weights θ^s. In particular, the teacher's weight θ_t^t is updated at step t as $\theta_t^t = \alpha\theta_{t-1}^t + (1-\alpha)\theta_t^s$, where α is the EMA decay. Therefore, the teacher gradually integrates information from different training steps through the EMA mechanism. This yields smooth and robust predictions, which then provide the student model with higher-quality pseudolabels. However, due to the numerous categories and complexity of RS images, the pseudolabels generated by the teacher often suffer from low quality and unreliability. Therefore, we purify them through a novel VLM-PP structure before providing the pseudolabels to the students.

B. Motivation for VLM-PP

In this section, we state the motivation and justification for proposing the VLM-PP for the SemiEarth model. The core challenge of RS S4 stems from the epistemic uncertainty inherent in pseudolabel generation. Unlike supervised learning, where ground truth provides unambiguous supervisory signals, pseudolabels are constrained by the knowledge of the teacher. However, the correctness of the predictions from the teacher cannot be guaranteed. The teacher model's erroneous predictions directly provide students with incorrect pseudolabels for learning, and these errors systematically propagate and amplify through iterative training. This creates a self-reinforcing loop where mistakes become increasingly entrenched rather than corrected. Existing approaches attempt to mitigate this issue through confidence-based thresholding, only retaining pseudolabels with a predefined confidence value. However, it is noteworthy that low-confidence regions are often located at ambiguous multiclass boundaries or at rare classes. Furthermore, it will make the model ineffective if we exclude these pixels from training. More importantly, existing methods cannot rectify the error pseudolabels, i.e., the traditional S4 structure cannot transcend the knowledge boundaries of its

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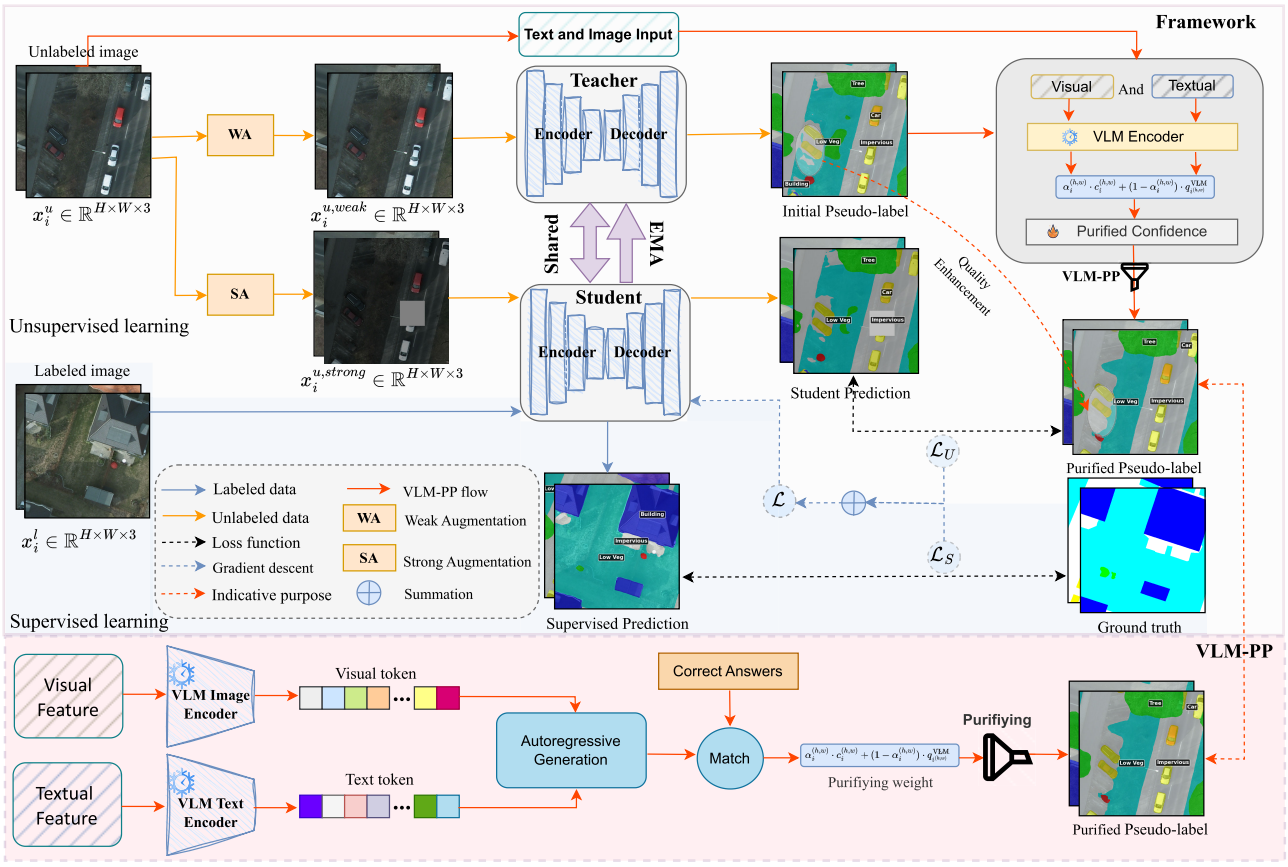


Fig. 2. Overall structure of our VLM-purified RS S4, i.e., SemiEarth, consists of unsupervised learning with unlabeled data and supervised learning with labeled data. The pseudolabels generated by the teacher are not directly provided to the student but first purified through the VLM-PP. Blue lines indicate the labeled data flow, yellow lines indicate the unlabeled, red lines indicate the VLM-PP flow, the blue dashed line indicates gradient descent and backpropagation, and the red dashed lines serve as an indicative purpose.

own learned features. Therefore, we introduce VLM-PP as external semantic knowledge to purify the pseudolabels.

We formulate the proposed purification procedure for logical reasoning and justification. First, we generate key pseudolabels and confidence scores as follows:

$$p_i^{u,t(h,w,k)} = P\left(pixel_{(h,w)} = k | x_i^{u,weak}\right) \tag{2}$$

$$y_i^{u,t(h,w)} = \arg\max_{k\in[1,K]} \left(p_i^{u,t(h,w,k)}\right) \tag{3}$$

$$c_i^{(h,w)} = \max_{k\in[1,K]} \left(p_i^{u,t(h,w,k)}\right) \tag{4}$$

where $x_i^{u,weak}$ represents the weakly augmented unlabeled data, (h, w) represents pixel coordinates, P is probability function, $k \in [1, K]$ is the class index, and $p_i^{u,t(h,w,k)}$ denotes the probability that the pixel at position (h, w) belonging to class k. For each pixel, the class with the highest probability is the pseudolabel $y_i^{u,t(h,w)}$. The maximum probability value $c_i^{(h,w)}$ is the confidence. The closer of $c_i^{(h,w)}$ to 1, the more certain the model is. Conversely, the closer to $1/K$, the more uncertain the model is, which means almost random. We define the low-confidence region as follows:

$$\mathcal{R}_{\text{low}} = \left\{(h, w) \mid c_i^{(h,w)} < \tau_{\text{conf}}\right\} \tag{5}$$

where τ_{conf} is the VLM-PP confidence threshold and $c_i^{(h,w)}$ represents the model's confidence scores in predicting the position (h, w), as calculated by (4). These low-confidence pixels, which the teacher model itself struggles to distinguish, are likely to further mislead the student model. Although early S4 methods attempted to filter out unreliable pseudolabels using confidence thresholds, simply discarding predictions below the threshold fails to address the problem, particularly in boundary regions. To address the issue of incorrect pseudolabels, it is essential to explore a novel architecture that is independent of conventional S4 methods. Our VLM-PP is a novel module that operates independently of the main S4 frameworks and can purify low-quality pseudolabels as well as correct errors in them.

C. VLM-PP Module

In this section, we introduce the scheme of VLM-PP for purifying pseudo-labels, as briefly illustrated in Fig. 3. The VLM-PP leverages the powerful visual understanding capabilities of VLMs for pseudolabel verification and purification. In short, VLMs combine visual information with text prompts to generate predictions, which are then compared with the correct answers for consistency evaluation.

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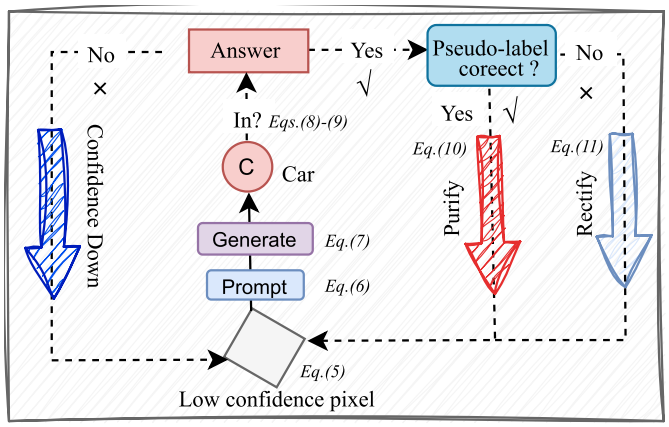


Fig. 3. Core logic of VLM purifying low-quality pseudolabels from teachers.

In particular, given unsupervised data x_i^u and a set of classes $\mathcal{N} = n_1, n_2, \ldots, n_K$, we construct a classification prompt $\mathcal{P}$ that enforces the model to identify all visible semantic categories as follows:

$$\mathcal{P} = \textit{List and locate all visible classes in the image.} \quad (6)$$

The VLM model generates a text prediction through autoregressive decoding by combining prompts with visual information. The specific process is defined as follows:

$$w_i^u = \arg\max_w \left(\prod_{t=1}^{T} P_\theta \left(w_t | w_{<t}, x_i^u, \mathcal{P} \right) \right) \quad (7)$$

where θ denotes the pretrained parameters of the generative VLM model, and T is the length of the output sequence. The term $P_\theta(w_t|w_{<t}, x_i^u, \mathcal{P})$ represents the conditional probability of generating a next word w_t given an image x_i^u, prompt $\mathcal{P}$, and previously generated words $w_{<t}$. w_i^u contains the category predicted by the VLM and the corresponding coordinates. To determine whether a candidate class is present in the generated sequence w_i^u and to obtain the VLM's confidence score, we apply the following criterion:

$$c_i^{u,k} = \begin{cases} \gamma, & \text{if class } n_k \text{ is mentioned in } w_i^u \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

For class n_k, $c_i^{u,k}$ is the confidence score from the VLM, and γ is a constant set to 0.95 in this article. For categories with nonzero confidence scores, we refine their pixel-level alignments using the SAM model. Finally, softmax normalization is applied to produce the final VLM confidence $\tilde{c}_i^{u,k}$

$$\tilde{c}_i^{u,k} = \frac{\exp\left(c_i^{u,k}\right)}{\sum_{j=1}^{K} \exp\left(c_i^{u,j}\right)}. \quad (9)$$

If the pseudolabel generated by the teacher model is of low confidence yet consistent with the VLM's predicted class, we perform pseudolabel purification as follows:

$$\tilde{c}_i^{(h,w)} = \alpha_i^{(h,w)} \cdot c_i^{(h,w)} + \left(1 - \alpha_i^{(h,w)}\right) \cdot \tilde{c}_i^{u,k(h,w)} \quad (10)$$

where $\alpha_i^{(h,w)} = c_i^{(h,w)}/\tau_{\text{conf}}$ is the purifying weight, $\tilde{c}_i^{u,k(h,w)}$ represents the VLM confidence score $\tilde{c}_i^{u,k}$ at the position (h, w), and $\tilde{c}_i^{(h,w)}$ is the final pseudolabel confidence. In regions where the teacher model is highly confident, we directly adopt its pseudolabels. In low-confidence regions, we perform pseudolabel purification to improve the reliability of unreliable pseudolabels via adaptive fusion with VLM-derived knowledge. Moreover, for pixels with low confidence, the parameter $(1 - \alpha_i^{(h,w)})$ in (10) increases, i.e., strengthening the purifying weight of VLM. Even for pseudolabels exhibiting extremely low confidence, VLM-PP can effectively purify them.

In addition, when the teacher-generated pseudolabel conflicts with the VLM's prediction, we replace it directly with the VLM's output, as shown in Fig. 3. For instance, if the teacher model provides a pseudolabel of a car while the VLM predicts a building, we consider it conflicting. Conversely, if the VLM also predicts a car, it is considered consistent. For these pixels, we adopt the VLM's confidence scores as the final pseudolabel confidence scores

$$\tilde{c}_i^{(h,w)} = \tilde{c}_i^{u,k(h,w)}. \quad (11)$$

Therefore, the VLM-PP can rectify misclassified pseudolabels produced by the teacher. Notably, we performed a secondary verification on the confidence scores of the purified pixels. Any purified pseudolabels that remain low-confidence are regarded as failed purification cases. Forcing the student model to learn from such pixels would be misleading. Therefore, these pixels are filtered out and excluded from the student model's training. The confidence threshold used for the secondary verification is set to 0.9, which is higher than that of the VLM-PP confidence threshold. This value, along with the VLM-PP confidence threshold mentioned earlier in this section, is discussed in detail in the ablation study Section IV-E7.

Algorithm 1 provides the core pseudocode for the SemiEarth training process, enhancing the interpretability and clarity of its methodology. The core logic of VLM-PP is shown in lines 8–15. D^U and D^L are the unsupervised and supervised datasets for one epoch, while N_U and N_L are the number of images in D^U and D^L. $model_s$ and $model_t$ denote the student and teacher models. $p_i^{u,s}$ denotes the student's final prediction on unlabeled data. $y^{u,t}$ represents the pseudo-labels generated by the teacher. $\mathcal{R}_{\text{low}}$ denotes the low-confidence region as defined in (5). $y_p^{u,t}$ denotes the purified pseudolabels. The CE is the Cross-Entropy loss function.

Our RS S4 model is the first to work to integrate VLMs into the RS S4 framework for pseudolabel purification. This section provides a thorough analysis of the motivation underlying the proposed VLM-PP model. In addition, we elaborate on the core principles of the VLM-PP module, an independent component that not only enhances pseudolabel quality but also enables the correction of erroneous pseudolabels. More importantly, SemiEarth departs from the conventional RS S4 architecture, offering strong interpretability while significantly outperforming state-of-the-art methods. It will be discussed in detail in Section IV.

IV. Experiments

In this section, we conduct experiments on RS datasets to validate our proposed novel RS S4 model. We first

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Algorithm 1 Training Procedure of SemiEarth

Input:
$D^U = \{(x_i^u)\}_{i=1}^{N_U}$, $D^L = \{(x_i^l, y_i)\}_{i=1}^{N_L}$
Output:
Θ: optimal model parameters

while until converge:
  for x_i^l, x_i^u in D^L, D^U:
    $\mathcal{L}_S = CE(model_s(x_i^l), y_i)$
    $x_i^{u,weak} = WeakAugment(x_i^u)$
    $x_i^{u,strong} = StrongAugment(x_i^u)$
    $p_i^{u,s} = model_s(x_i^{u,strong})$
    $y^{u,t} = model_t(x_i^{u,weak})$
    for pixels in $y^{u,t}$:
      if pixels in $\mathcal{R}_{\text{low}}$:
        Confidence estimation via VLM by Eq. 7-9
        if pseudo-label matches VLM prediction:
          Purify pseudo-labels via Eq. 10.
        else:
          Rectify with VLM Pseudo-Labels.
      if pixels' confidence is still low:
        Filter out and exclude pixels.
    end for
    $\mathcal{L}_U = CE(p_i^{u,s}, y_p^{u,t})$
    $\mathcal{L} = \mathcal{L}_S + \mathcal{L}_U$
    Update Θ via gradient descent on $\mathcal{L}$
    Save the best checkpoint Θ_{best}
  end for
return Θ
end

conduct quantitative and qualitative comparisons with SOTA models. Then, we perform ablation studies on SemiEarth to validate the effectiveness of the proposed modules. Our experimental code, dataset split files, and detailed experimental setup are all released at https://github.com/wangshanwen001/SemiEarth. In addition, the processed datasets used in our paper and our model checkpoints can be downloaded via https://huggingface.co/papers/2602.00202

A. RS Datasets and Data Augmentation

1) LoveDA: The LoveDA RS dataset [55] consists of 5987 images containing a total of 166 768 annotated objects, collected from three distinct urban regions. Each image has a spatial resolution of 0.3 m and a size of 1024 × 1024 pixels, and it is labeled with one of seven semantic classes: building, road, water, barren, forest, agriculture, and background. Due to the GPU memory limitations, all images are resized and cropped to 512 × 512 pixels, yielding 16 764 training samples for deep learning. Following the protocol of the previous SOTA work on RS S4 [17], using a 6:2:2 split into training, validation, and test sets for local evaluation, without submission to the dataset's competition website. The data we ultimately used to validate the model's performance is completely isolated from the training data. The specific processed datasets and data splits can be found in our open-source repository.

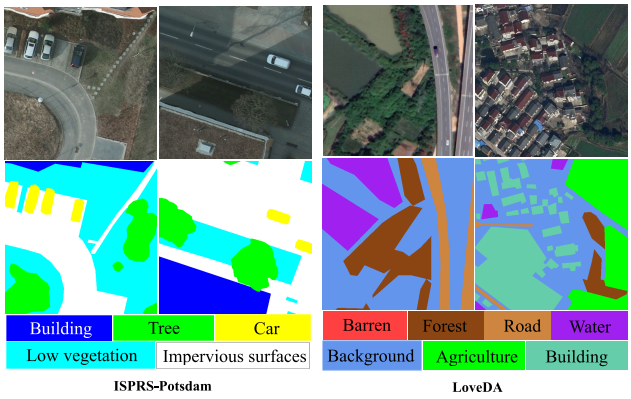


Fig. 4. Samples from the two RS Datasets.

2) ISPRS-Potsdam: The ISPRS Potsdam RS dataset [56] is widely used to advance research in semantic segmentation of the RS imagery. It has a resolution of 0.05 m and consists of 38 super-large satellite RS images, each with a size of 6000 × 6000 pixels. The dataset contains six segmentable classes, i.e., impervious surfaces, building, low vegetation, tree, car, and background. We crop each original image into 512 × 512 patches, resulting in a total of 5472 cropped images for deep training. Consistent with the LoveDA dataset, we adopted the same data processing approach as the previous RS S4 SOTA, splitting the data into training, validation, and test sets in a 6:2:2 ratio. The data used to validate the model's performance is completely isolated from the training data.

Some samples from these RS datasets are shown in Fig. 4, where categories are numerous, and boundaries among multiple classes are complex. We employed both weak and strong augmentations on the datasets. On unlabeled images, weak augmentation was performed with geometric transformations such as image scaling, horizontal flipping, and vertical flipping, while strong augmentation was implemented using methods such as photometric transformations, Gaussian blur, and CutMix [57].

B. Evaluation Metric and Experimental Setup

Following the standard evaluation protocol of the previous RS S4 methods, we use mIoU as the primary metric to assess model performance. The mIoU is calculated as the average IoU across all classes

$$IoU_k = \frac{TP_k}{TP_k + FP_k + FN_k} \tag{12}$$

$$mIoU = \frac{1}{K}\sum_{k=1}^{K} IoU_k \tag{13}$$

where TP_k, FP_k, and FN_k represent true positives, false positives, and false negatives for class k, and K is the total number of classes. To evaluate the model's theoretical computational speed and actual inference speed, we incorporate the *FLOPs* and *FPS* metrics, with their definitions detailed as follows:

$$FLOPs = 2 \times H \times W \times \left(C_{in}K^2\right) \times C_{out} \tag{14}$$

$$FPS = \frac{1}{\text{Inference time (s)}} \tag{15}$$

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where C_{in} represents the number of input channels, K stands for the number of convolution kernels, and C_{out} indicates the number of output channels. The *FLOPs* measure the theoretical number of multiply-accumulate operations in a model, with lower values being better. The *FPS* represents the actual inference speed, with higher values indicating faster model inference.

Our model is implemented in PyTorch and trained on eight NVIDIA RTX 4090 GPUs and one NVIDIA RTX H100. We adopt DINOv2_small [64] as the teacher–student backbone network and utilize Qwen-VL-3B [65], [66] as the VLM. SemiEarth is trained for 50 and 20 epochs on the Potsdam and LoveDA datasets, respectively. Notably, previous RS S4 models also exclude the clutter class in Potsdam and the Ignore class in LoveDA when calculating the final mIoU [11], [26], [29], [61]. To ensure consistency with prior work, we adopt the same evaluation protocol. Further implementation and experimental details can be found in our open-source repository.

C. Quantitative Results Compared to SOTA

This section conducts experiments on ISPRS-Potsdam and LoveDA datasets, compared to the SOTA methods, including mean teacher [58], CutMix [57], CCT [59], CPS [60], LSST [61], FixMatch [14], UniMatch [62], DWL [26], Allspark [63], MUCA [11], SemiVL [23], and RS-MTDF [24]. In particular, we show the results for labeled data ratios of 1%, 5%, and 10% to verify the effectiveness, respectively. Notably, for the previous SOTA methods, we adopt the network configurations as default in their papers and code, with some records directly referenced from their original papers.

The experimental results on the ISPRS-Potsdam and LoveDA datasets are shown in Tables I and II, respectively. In Table II, "Bac" and "Agr" are abbreviations for the background and agriculture classes, respectively. We can observe that S4 methods, such as mean teacher and CPS, achieve the poorest results. On the contrary, S4 models like UniMatch and AllSpark yield relatively good results. This improvement is attributed to their careful filtering or weighting of pseudolabels in the unlabeled data, which reduces the harmful impact of low-confidence samples. The DWL and MUCA devise domain-specialized architectures that significantly boost performance by addressing the unique characteristics of RS data, such as rich multiscale information and high interclass similarities. Moreover, SemiVL and RS-MTDF achieve a higher mIoU than traditional S4 methods; they still fall short of our SemiEarth. It is worth noting that the RS-MTDF here uses DINOv2_small as the backbone. A more detailed comparison between our SemiEarth and this model is provided in Section IV-E5 of the ablation study. From the results, we can see that the proposed SemiEarth achieves the best mean mIoU across all classes on both RS datasets.

In addition, SemiEarth not only achieves SOTA performance in quantitative comparisons with existing models but also marks a substantial step forward for RS S4 methods. In addition, on the ISPRS-Potsdam dataset, with labeled data ratios of 1%, 5%, and 10%, our model outperforms the second-best model, MUCA, by 5.69%, 4.39%, and 5.13% in mIoU, respectively. Similarly, on the LoveDA dataset under 1%, 5%, and 10% labeling ratios, our model dramatically outperforms all existing methods, surpassing the MUCA model by substantial margins of 6.47%, 10.37%, and 11.0% in mIoU, respectively. Moreover, even though SemiVL and RS-MTDF utilize VLM or VFM models, their mIoU remains lower than that of SemiEarth. This is because, although earlier S4 methods filtered out low-confidence pseudolabels, such regions in RS images often correspond to class boundaries with high semantic ambiguity. Excluding pixels from these low-confidence regions during training compromises model performance. On the contrary, SemiEarth introduces a novel purification strategy for low-confidence pixels, effectively addressing this limitation.

TABLE I
COMPARISON RESULTS WITH SOTA METHODS ON ISPRS-POTSDAM DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. IOU AND MIOU ARE REPRESENTED AS PERCENTAGES

Ratio	Model	IoU					mIoU
		Building	Low vegetation	Tree	Car	Impervious surfaces	
1%	Mean teacher [58]	72.53	58.98	60.43	67.97	67.39	65.46
	CutMix [57]	55.58	42.05	49.72	50.86	39.40	47.52
	CCT [59]	54.48	61.28	48.56	52.95	60.71	55.59
	CPS [60]	59.35	69.16	62.89	59.88	66.33	63.52
	LSST [61]	68.74	75.24	54.74	62.09	68.80	65.92
	FixMatch [14]	76.95	71.59	64.71	65.85	72.81	70.38
	UniMatch [62]	76.52	70.99	65.44	66.62	72.64	70.44
	DWL [26]	72.34	**77.08**	62.74	62.57	72.22	69.39
	AllSpark [63]	83.70	65.92	59.64	69.77	75.31	70.87
	MUCA [11]	84.56	66.98	56.96	71.52	76.64	71.33
	SemiVL [23]	84.73	67.28	58.87	72.95	77.16	72.20
	RS-MTDF [24]	85.14	68.59	67.44	71.77	76.35	73.86
	Our (SemiEarth)	**86.80**	71.22	**71.96**	**76.11**	**79.01**	**77.02**
5%	Mean teacher [58]	82.15	65.92	67.11	72.21	74.60	72.40
	CutMix [57]	52.94	68.86	41.51	58.33	54.82	55.29
	CCT [59]	72.90	80.25	64.23	58.32	74.42	70.02
	CPS [60]	76.53	84.34	57.98	69.45	75.39	72.74
	LSST [61]	69.26	84.55	67.33	67.49	73.86	72.50
	FixMatch [14]	78.12	74.87	68.89	66.58	75.30	72.75
	UniMatch [62]	78.24	73.59	67.17	66.64	75.07	72.14
	DWL [26]	74.81	**85.64**	66.38	62.99	75.68	73.10
	AllSpark [63]	85.57	67.62	60.61	73.48	77.15	72.88
	MUCA [11]	88.45	69.53	61.39	74.18	79.56	74.62
	SemiVL [23]	87.58	70.37	63.59	75.06	78.92	75.10
	RS-MTDF [24]	87.66	72.31	71.04	76.45	76.99	76.89
	Our (SemiEarth)	**88.51**	74.45	**74.06**	**78.14**	**79.87**	**79.01**
10%	Mean teacher [58]	84.76	69.28	68.83	71.66	76.51	74.21
	CutMix [57]	64.55	80.99	64.79	65.50	68.01	68.77
	CCT [59]	73.09	83.94	61.12	60.45	73.06	70.33
	CPS [60]	77.80	87.15	61.12	68.48	75.89	74.09
	LSST [61]	70.92	86.06	68.91	70.22	74.89	74.20
	FixMatch [14]	77.97	76.17	70.09	70.97	76.14	74.27
	UniMatch [62]	77.34	87.75	70.79	56.65	76.46	73.80
	DWL [26]	76.37	**88.42**	66.54	64.37	77.14	74.57
	AllSpark [63]	86.29	69.83	64.17	75.23	78.31	74.76
	MUCA [11]	88.02	70.58	64.53	75.20	79.92	75.65
	SemiVL [23]	88.62	72.23	65.40	76.34	80.71	76.66
	RS-MTDF [24]	88.57	73.11	71.35	76.53	78.91	77.69
	Our (SemiEarth)	**90.59**	75.44	**75.01**	**79.64**	**83.24**	**80.78**

D. Qualitative Results Compared to SOTA

To better illustrate the advantages of SemiEarth over other SOTA models, we provide a visual comparison of semantic segmentation results on RS datasets. Figs. 5 and 6 present qualitative comparisons with SOTA methods on the Potsdam and LoveDA datasets, respectively. For clarity, we highlight regions prone to missegmentation with dashed ellipses,

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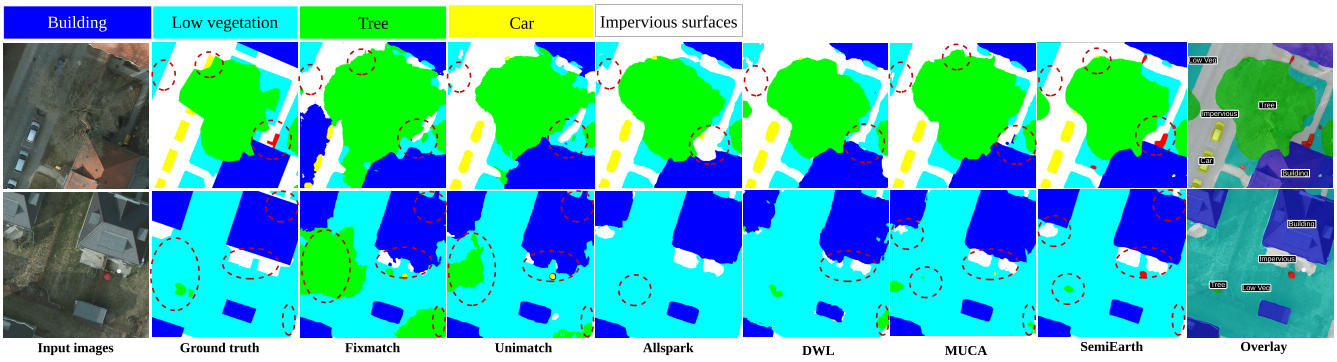


Fig. 5. Qualitative results with different SOTA S4 methods on the ISPRS-Potsdam dataset.

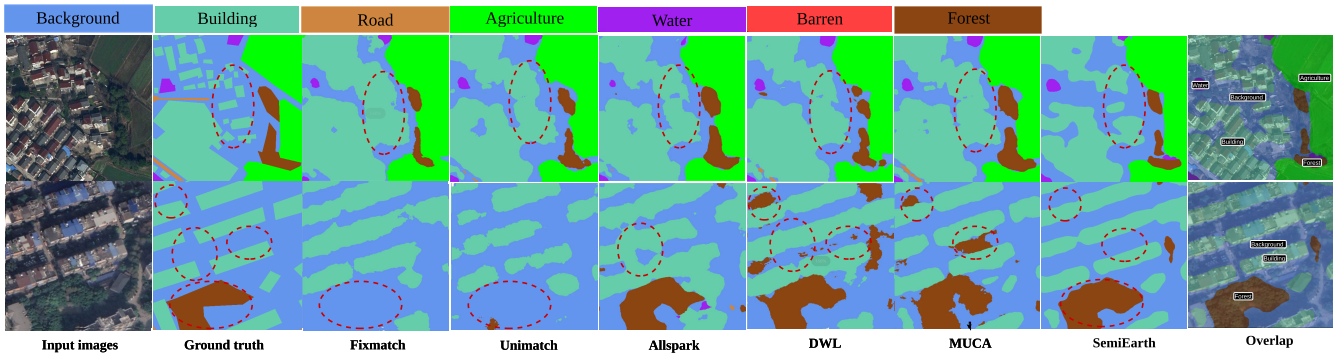


Fig. 6. Qualitative results with different SOTA S4 methods on the LoveDA dataset.

emphasizing the comparatively accurate segmentation achieved by our model.

On the ISPRS Potsdam dataset, traditional S4 methods exhibit numerous misclassifications, particularly in complex scenes with complex and overlapping categories. Fixmatch and Unimatch make errors in segmenting regions for the low vegetation, tree, impervious surfaces, and building classes. The RS S4 models, such as DWL and MUCA, achieve promising performance in these regions, but they performed poorly at the boundaries of multiple categories. In comparison, SemiEarth achieves the highest segmentation accuracy among all evaluated methods. In addition, we overlay our segmentation results on the original images and provide corresponding visual labels in the last column to facilitate comparison. The overlays demonstrate that our model produces highly accurate predictions, particularly along the boundaries between multiple semantic classes. Notably, the small red region in the first row of the ground truth corresponds to the clutter class defined in the official ISPRS-Potsdam dataset. However, due to its minimal presence in the scene, prior RS S4 methods excluded this class from quantitative evaluation [11], [25], [26], and also failed to segment it accurately in qualitative results. On the contrary, SemiEarth correctly segments this challenging region.

On the LoveDA dataset, all the compared methods exhibit degraded visual quality. In particular, complex regions with intermixed classes suffer from segmentation errors in most approaches. Moreover, for categories requiring fine-grained segmentation, such as the boundaries of isolated buildings and forest areas (see Fig. 6), most S4 models fail to produce accurate segmentation. On the contrary, SemiEarth produces more accurate segmentations for classes including building, forest, agriculture, and background. To facilitate visual comparison, we overlay our segmentation results on the original images and provide class labels in the last column. These overlays demonstrate that our model yields higher-quality results on the LoveDA dataset, with improved boundary segmentation and fewer misclassifications compared to existing approaches.

E. Ablation Study

In this section, we conduct detailed ablation experiments on SemiEarth to validate the rationality of our model.

1) Ablation Study of VLM-PP: We perform ablation studies on the core component VLM-PP of the SemiEarth model, i.e., without the VLM-PP, with only purification, with only rectification, and with the full VLM-PP. In this section and the following ones, both the teacher and student networks employ DINOv2_small as the backbone and Qwen-VL-3B as VLM, with a labeled data ratio of 5%.

The experimental results are shown in Table III. It can be seen that without the proposed core component, VLM-PP, the performance is low. Meanwhile, after incorporating the purification and rectification components of VLM-PP, the mIoU shows an increase compared to the baseline. In particular, on the Potsdam dataset, the purification component and rectification achieved improvements of 3.74% and 2.37% over the baseline, respectively, and with gains of 3.85% and

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TABLE II

COMPARISON RESULTS WITH OTHER SOTA METHODS ON THE LOVEDA DATASET. THE BEST RESULTS ARE IN BOLD. IOU AND MIOU ARE REPRESENTED AS PERCENTAGES

Ratio	Model	IoU							mIoU
		Bac	Building	Road	Water	Barren	Forest	Agr	
1%	Mean teacher [58]	44.73	42.53	40.34	50.92	11.37	26.88	54.40	38.73
	CutMix [57]	36.04	24.69	10.03	24.60	3.43	6.67	10.19	16.52
	CCT [59]	37.16	22.41	27.86	43.98	14.51	25.38	36.67	29.71
	CPS [60]	46.52	20.87	27.85	50.55	0.01	33.16	34.60	30.51
	LSST [61]	44.73	41.90	39.90	62.65	29.27	31.26	48.29	42.57
	FixMatch [14]	46.78	51.20	50.21	67.27	11.53	**36.79**	50.26	44.86
	UniMatch [62]	46.53	51.38	49.36	**67.74**	10.86	33.40	52.28	44.51
	DWL [26]	48.74	56.79	51.59	63.42	22.56	35.20	55.38	47.67
	AllSpark [63]	63.87	47.70	46.05	61.52	35.31	30.94	55.64	48.72
	MUCA [11]	64.89	56.03	47.14	63.86	35.81	22.57	58.18	49.78
	SemiVL [23]	64.26	56.05	50.97	60.83	49.19	22.42	56.31	51.43
	RS-MTDF [23]	50.50	56.71	51.74	63.73	52.21	35.42	57.26	52.51
	Our (SemiEarth)	**64.99**	**57.12**	**56.89**	56.86	**73.17**	25.69	**59.03**	**56.25**
5%	Mean teacher [58]	49.73	46.22	42.34	60.93	31.51	35.79	44.22	44.39
	CutMix [57]	41.48	41.62	38.77	47.44	14.69	28.09	31.05	34.73
	CCT [59]	46.80	44.62	46.80	60.95	24.83	29.03	44.30	42.48
	CPS [60]	48.90	49.64	47.97	60.27	4.67	36.09	47.32	42.12
	LSST [61]	51.48	45.66	52.66	67.63	33.52	35.80	48.60	47.91
	FixMatch [14]	45.40	53.05	51.22	66.73	28.53	27.25	54.30	44.64
	UniMatch [62]	50.20	54.49	50.46	67.18	26.79	30.06	54.86	47.72
	DWL [26]	48.75	55.00	51.53	**69.49**	29.46	36.59	52.11	48.99
	AllSpark [63]	65.09	55.06	47.59	67.10	34.67	26.86	51.87	49.75
	MUCA [11]	67.29	56.04	48.37	61.02	36.21	30.76	57.09	50.97
	SemiVL [23]	66.34	56.64	51.58	61.01	51.25	25.77	57.86	52.92
	RS-MTDF [23]	63.55	57.30	52.69	64.40	59.39	39.99	58.07	56.48
	Our (SemiEarth)	**69.57**	**57.70**	**62.12**	60.52	**75.61**	**42.88**	**60.98**	**61.34**
10%	Mean teacher [58]	50.45	55.75	43.56	66.15	35.24	36.96	45.64	47.68
	CutMix [57]	46.73	49.60	47.36	59.99	29.06	37.77	40.60	44.44
	CCT [59]	44.07	45.22	47.65	57.12	24.41	32.50	45.07	42.29
	CPS [60]	51.30	54.93	52.57	53.37	18.39	37.59	53.24	45.91
	LSST [61]	50.69	49.50	52.63	69.85	27.25	36.24	52.06	48.32
	FixMatch [14]	52.02	55.59	53.20	57.91	25.86	40.83	57.50	48.99
	UniMatch [62]	51.80	53.95	51.17	58.15	25.60	38.72	54.86	47.75
	DWL [26]	49.94	56.66	53.89	**70.35**	30.62	41.49	53.13	50.87
	AllSpark [63]	67.13	56.16	40.67	63.58	32.54	32.03	56.91	49.86
	MUCA [11]	68.69	58.20	41.82	65.62	37.09	35.01	57.38	51.97
	SemiVL [23]	67.29	56.82	52.38	61.31	52.17	28.56	58.54	53.87
	RS-MTDF [23]	64.06	57.41	55.83	65.62	65.95	41.18	58.75	58.40
	Our (SemiEarth)	**71.10**	**60.24**	**64.87**	62.44	**76.70**	**42.42**	**63.05**	**62.97**

TABLE III

ABLATION OF THE VLM-PP

Dataset	Network	mIoU
ISPRS-Potsdam	Without VLM-PP	74.42
	Only Purification	78.16
	Only Rectification	76.79
	Full VLM-PP	79.01
LoveDA	Without VLM-PP	56.33
	Only Purification	60.18
	Only Rectification	58.85
	Full VLM-PP	61.34

2.52% on LoveDA. Moreover, the performance improves significantly with VLM-PP. In particular, the mIoUs are increased by 4.59% and 5.01% on the Potsdam and LoveDA RS datasets, respectively. The experimental results demonstrate that our proposed VLM-PP significantly enhances the performance of the S4 model in the RS domain.

2) Visualized Results of the VLM-PP Module: To better illustrate the effectiveness of the VLM-PP module, we visualize the pseudolabels and their confidence scores generated by the teacher model—both before and after applying VLM-PP within the same training iteration.

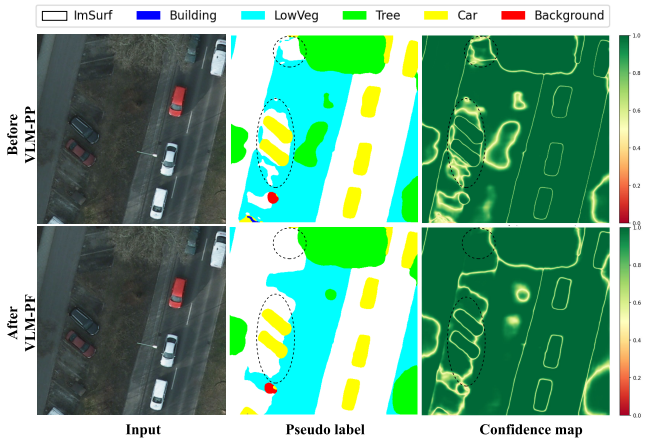


Fig. 7. Before and after the VLM-PP module: comparison of the quality of pseudo-labels and confidence.

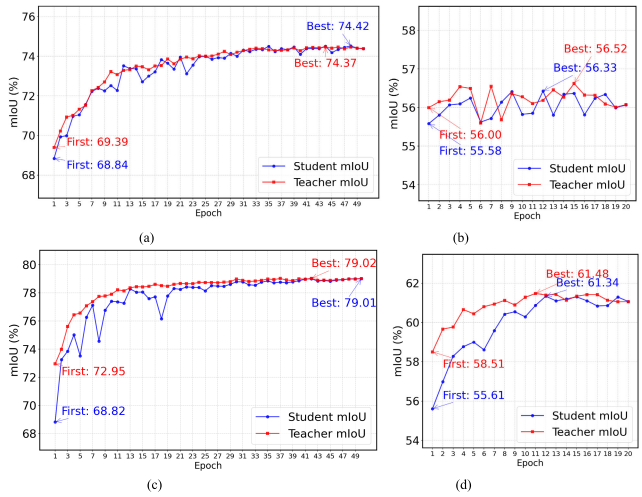


Fig. 8. VLM-PP significantly improves the quality of pseudolabels generated by the teacher model, enabling it to effectively guide the student model during training iterations. (a) Potsdam without VLM-PP. (b) LoveDA without VLM-PP. (c) Potsdam with VLM-PP. (d) LoveDA with VLM-PP.

The results are shown in Fig. 7. In complex regions of RS images, particularly at boundaries among low vegetation, car, tree, and impervious surfaces, the pseudolabels generated by the teacher network before VLM-PP are spatially fragmented. Moreover, their associated confidence scores are consistently low, reflecting the teacher's uncertainty in assigning labels to these multiclass boundary zones. Such unreliable supervision can mislead the student model into learning erroneous patterns. On the contrary, after applying the VLM-PP module, the teacher produces pseudolabels with significantly higher confidence and improved semantic consistency in these challenging boundary zones. This is because VLM-PP not only reinforces correctly predicted regions by boosting their confidence but also rectifies misclassified pixels, thereby yielding more accurate and reliable pseudolabels for student training.

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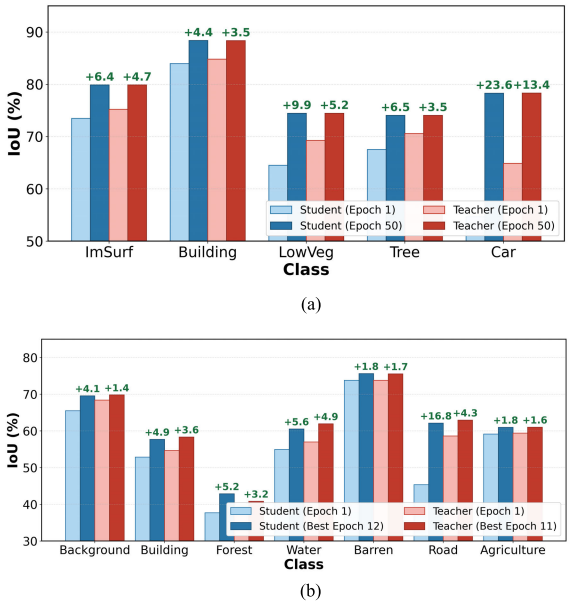


Fig. 9. Improvement in IoU from the start to the end of training for each class on the RS datasets. Initial and final IoU for each class in (a) Potsdam and (b) LoveDA.

3) Purified Pseudolabels Improve Student Learning: In this section, we visualize how the teacher network progressively guides the student network with purified pseudolabels throughout the training process. The teacher model generates pseudolabels that serve as supervision for the student, and we assess the quality of these pseudo-labels using the mIoU metric. Fig. 8 compares the SemiEarth iteration dynamics under two settings: with and without VLM-PP purification.

At the beginning of training, the performance of the student is nearly identical with and without VLM-PP. Then, the quality of pseudolabels from the teacher becomes crucial for effective learning. Further analysis reveals that the segmentation accuracy of the teacher without VLM-PP is only marginally better than (or even comparable to) that of the student. Therefore, the teacher offers little effective supervision to the student. Moreover, in the absence of VLM-PP, both models rapidly overfit the limited labeled data as training progresses. Their learning dynamics become tightly coupled because the parameters of the teacher are updated via EMA from the student and lack a mechanism to refine low-quality pseudolabels. Once the teacher can no longer generate pseudolabels that are more accurate than the student's own predictions, their performance fluctuates in lockstep, showing no consistent improvement.

On the contrary, with VLM-PP enabled, the teacher provides consistently high-quality supervision from the early stages of training. Its pseudolabels outperform the student's own predictions, enabling the student to progressively learn accurate semantic representations. This advantage stems from VLM-PP's ability to purify the teacher's pseudolabels.

In addition, as illustrated in Fig. 9, we visually present the perclass IoU improvements of our model on the RS datasets over the course of training. In both datasets, the student model exhibits the most significant gains for the Car and Road classes, with IoU increases of 23.6% and 16.8%, respectively. These categories are particularly challenging, e.g., a car occupies only a small fraction of pixels in RS images, while the road has irregular and ambiguous boundaries. The strong performance on these difficult classes can be attributed to the high-quality pseudolabels consistently generated by the teacher model throughout training.

4) Ablation Study of VLM Inference Frequency: In this section, we investigate the impact of VLM call frequency by varying the interval from every iteration to every 100 iterations. Notably, an "iteration" here refers to a single optimization step, distinct from an "epoch," which denotes a complete pass through the entire training dataset.

The experimental results are shown in Table IV. The FLOPs represent the theoretical computational cost of a single forward pass, which is inherently independent of how frequently the VLM is called. However, when programmatically calculating FLOPs, the VLM is typically bundled into the model's total FLOPs. To provide a realistic estimation of the computational workload during training, we compute the average FLOPs over a 100-iteration window. For example, if the VLM is invoked once every ten iterations, we calculate the FLOPs for ten iterations with the VLM and 90 iterations without, and then take the average. The results indicate that the average training FLOPs steadily rise as the VLM invocation frequency increases. During evaluation, neither the teacher model nor the VLM is active; thus, the inference FLOPs depend solely on the student model and remain entirely unaffected by the VLM calling frequency.

Regarding training speed, the FPS decreases nonlinearly as the VLM invocation frequency increases. In particular, on the Potsdam dataset, invoking the VLM every 100 iterations yields 47.7 frames/s, whereas invoking it every ten iterations results in 44.2 frames/s. Thus, a ten-fold increase in invocation frequency incurs a marginal drop of only 3.5 frames/s, while the mIoU gains a substantial 2.01%. This is because the ample training iterations between VLM calls allow the GPU to maintain high throughput, preventing a severe performance drop. Concurrently, the higher invocation frequency significantly improves pseudolabel quality. As the VLM inference interval decreases from every ten iterations to every 2, the frames/s decline gradually. However, a sharp drop occurs when reducing the interval from every two iterations to every single iteration. We attribute this to the fact that at intervals greater than 2, the FPS is primarily constrained by scheduling overhead, leaving the GPU with idle capacity for data transfer. Conversely, at an interval of 1, the combined effects of cache invalidation and continuous inference during training saturate our laboratory GPU's limits, causing a severe FPS drop. This bottleneck stems from current hardware limitations and could likely be mitigated with a more powerful GPU configuration. Consequently, we select an invocation interval of two iterations as the optimal tradeoff between mIoU and training speed for our setup. Given superior hardware, invoking the VLM at every iteration would be viable. Note that during evaluation, since neither the teacher model nor the VLM is involved, the inference FPS depends solely on the trained student model and remains unaffected by the training-time VLM frequency.

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TABLE IV
ABLATION STUDY OF VLM INFERENCE FREQUENCY

Dataset	Iteration frequency	mIoU	Training FPS	Training FLOPs	Evaluation FPS	Evaluation FLOPs
ISPRS-Potsdam	every iteration	79.98	0.8	242.82G	162.0	80.94G
	every 2 iterations	79.01	25.3	202.35G	162.0	80.94G
	every 5 iterations	78.08	40.6	182.12G	162.0	80.94G
	every 10 iterations	77.07	44.2	169.97G	162.0	80.94G
	every 100 iterations	75.06	47.7	162.69G	162.0	80.94G
LoveDA	every iteration	61.93	0.7	242.89G	160.8	80.96G
	every 2 iterations	61.34	24.5	202.41G	160.8	80.96G
	every 5 iterations	60.60	38.6	182.16G	160.8	80.96G
	every 10 iterations	58.89	40.5	170.01G	160.8	80.96G
	every 100 iterations	56.87	44.1	162.73G	160.8	80.96G

TABLE V
ABLATION STUDY OF BACKBONE AND COMPARE TO RS-MTDF

Dataset	Method	mIoU	Training FPS	Training FLOPs	Evaluation FPS	Evaluation FLOPs	Parameter	VLM or VFM Parameter
ISPRS-Potsdam	RS-MTDF (DINOv2_small)	76.89	22.1	228.26G	146.9	85.70G	25.2M	430M
	SemiEarth (DINOv2_small)	79.01	25.3	202.35G	162.0	80.94G	24.8M	3B
	RS-MTDF (DINOv2_base)	81.62	13.2	850.49G	54.3	343.63G	98.0M	430M
	SemiEarth (DINOv2_base)	82.38	14.6	805.40G	65.2	322.16G	97.5M	3B
LoveDA	RS-MTDF (DINOv2_small)	56.48	21.9	228.33G	144.8	85.75G	25.2M	430M
	SemiEarth (DINOv2_small)	61.34	24.5	202.41G	160.8	80.96G	24.8M	3B
	RS-MTDF (DINOv2_base)	61.57	13.0	850.73G	51.8	343.82G	98.0M	430M
	SemiEarth (DINOv2_base)	63.08	14.3	805.61G	61.9	322.33G	97.5M	3B

5) Ablation Study of Backbone and Compare to RS-MTDF: Since the VLM remains frozen to refine pseudolabels without undergoing parameter updates during SemiEarth training, only the teacher and student models are actively trained. Consequently, we report the backbone and VLM parameters separately. While our previous framework utilized DINOv2_small as the backbone for both teacher and student networks, we have now introduced DINOv2_base as an additional backbone. We evaluated the impact of these different backbone networks on both parameter count and mIoU. To ensure a fair comparison, we benchmarked our approach against RS-MTDF [24] using the same backbone. We implemented a DINOv2_small version of RS-MTDF and conducted all experiments under identical environmental conditions and configurations.

As presented in Table V, our method outperforms RS-MTDF in both mIoU and inference speed under identical setups. In particular, when utilizing the same backbone, RS-MTDF has a slightly larger parameter count than SemiEarth due to architectural differences. Furthermore, although RS-MTDF employs CLIP, which possesses fewer parameters than our VLM-PP, its final FLOPs remain slightly higher than those of SemiEarth. This efficiency stems from variations between our network and RS-MTDF design, as well as the fact that our VLM is not invoked at every iteration. Consequently, the inference FPS of RS-MTDF is slightly lower than that of SemiEarth. Note that the actual FPS is highly dependent on the GPU environment. Our reported FPS reflects the speed achieved under our specific hardware setup.

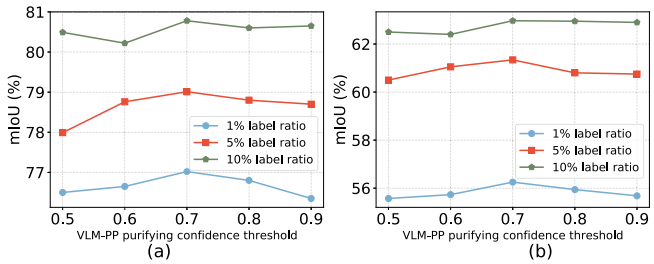


Fig. 10. Values of the different purifying confidence thresholds. (a) Potsdam. (b) LoveDA.

6) Purification Confidence Threshold in VLM-Pp: We investigate the optimal confidence threshold for VLM-PP to purify pseudolabels generated by the teacher model. In particular, we evaluate thresholds ranging from 0.5 to 0.9, with results presented in Fig. 10. As the purification threshold increases, the model's mIoU initially rises and then declines. This trend can be explained as follows: a threshold that is too low fails to filter out low-quality pseudolabels, thereby introducing noise that misleads the student model; conversely, a threshold that is too high discards even highly confident predictions

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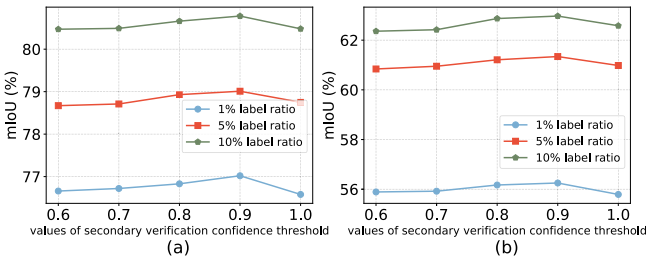


Fig. 11. Values of the different secondary verification confidence thresholds. (a) Potsdam. (b) LoveDA.

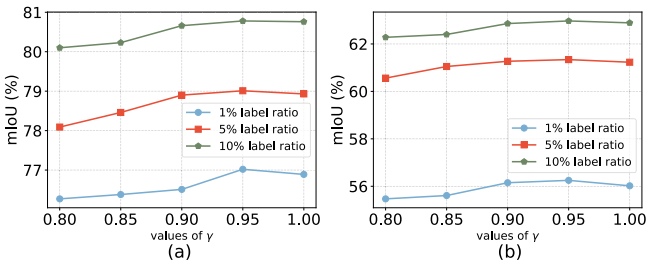


Fig. 12. Values of the different γ. (a) Potsdam. (b) LoveDA.

from the teacher, unnecessarily reducing the amount of useful supervision. The best performance is achieved within the range of 0.7–0.8, and we adopt 0.7 as the final threshold for all comparative experiments.

This section validates the effectiveness and necessity of the proposed components through systematic ablation studies. The VLM-PP module significantly enhances performance on RS S4, while its underlying mechanism provides strong interpretability.

7) Ablation Study of Secondary Verification Confidence Threshold: We investigate the optimal secondary confidence threshold for purified pseudolabels. Since these pixels have already been purified, our secondary verification confidence threshold is stricter than the VLM-PP purifying confidence threshold. In particular, we evaluate thresholds ranging from 0.6 to 1, with results presented in Fig. 11. As the purification threshold increases, the model's mIoU initially rises and then declines, peaking at 0.9. This trend can be explained as follows: Since the pseudolabels have already been purified, an overly low secondary verification confidence threshold lacks sufficient discriminative power, resulting in suboptimal performance. Conversely, setting the threshold to one is excessively strict, which filters out many pseudolabeled pixels with confidence scores above 0.9. Therefore, a threshold of around 0.9 yields the best results.

8) Ablation Study of γ: In this section, we investigate the optimal value of γ. In particular, we evaluate thresholds ranging from 0.8 to 1, with results presented in Fig. 12. As the γ value increases, the model's mIoU gradually rises and peaks at 0.95, followed by a slight decline. Therefore, we ultimately selected 0.95.

V. Conclusion

Our study addresses the challenge of low-quality pseudolabels associated with semisupervised RS image semantic segmentation by proposing the SemiEarth model. SemiEarth adopts an overall teacher–student model architecture and incorporates a specialized pseudolabel purification module named VLM-PP. The VLM-PP, as a novel and independent module, aims to purify the low-quality pseudolabels provided by the teacher model, preventing the student model from being misled. Compared to other state-of-the-art methods, our model achieves the highest mIoU on RS datasets and, unlike most prior works, provides strong interpretability. In particular, detailed ablation studies demonstrate that VLM-PP consistently and reliably purifies low-quality pseudolabels across all training iterations, thereby boosting the student model's learning performance. Nevertheless, current semisupervised methods for RS image analysis still encounter several key bottlenecks. For instance, while this article pioneers the integration of VLMs into the RS S4 domain, the potential of combining large AI models with complex spectral RS imaging, e.g., hyperspectral data, within the S4 framework remains largely unexplored. Our future work will focus on investigating how VLMs can be effectively leveraged for such challenging, complex spectral RS imaging in S4. Finally, as the first work to introduce VLMs into the RS S4 domain, we hope that SemiEarth will serve as a valuable benchmark and help catalyze future research in this emerging direction.

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